

HUAWEI CLOUD-NATIVE

AIOps, Observability & Automated Incident Response

An evidence-driven SRE operating model for CCE

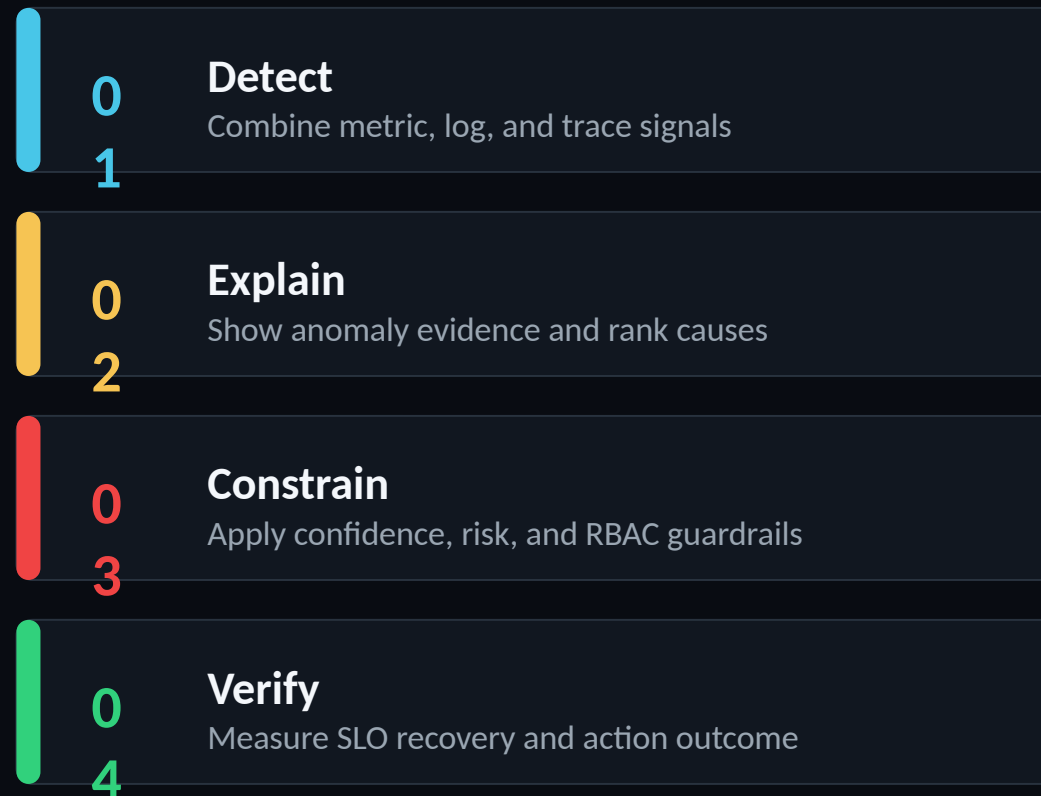
Murat Miraç Gedik

Portfolio Reference Architecture • 2026

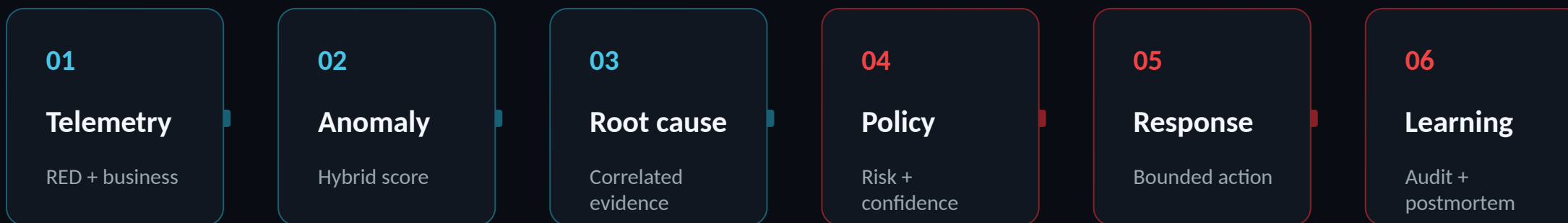


Seeing failure is not enough; response must be safe and correct

Fragmented telemetry, noisy alerts, and manual recovery extend detection time and increase the risk of the wrong action.

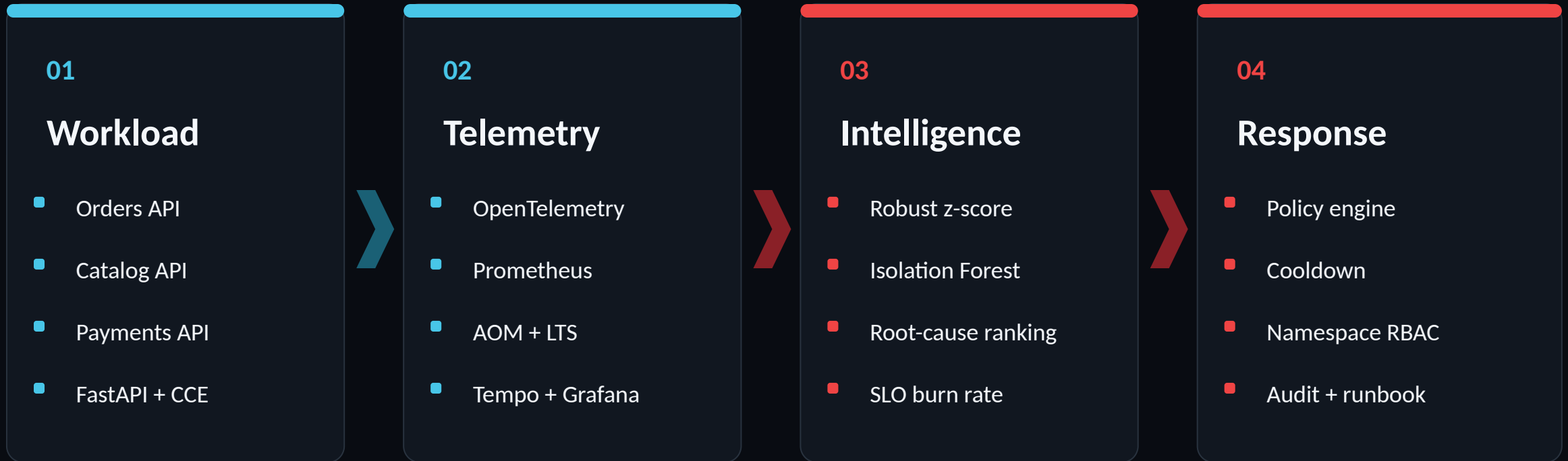


The platform turns the incident lifecycle into one evidence chain

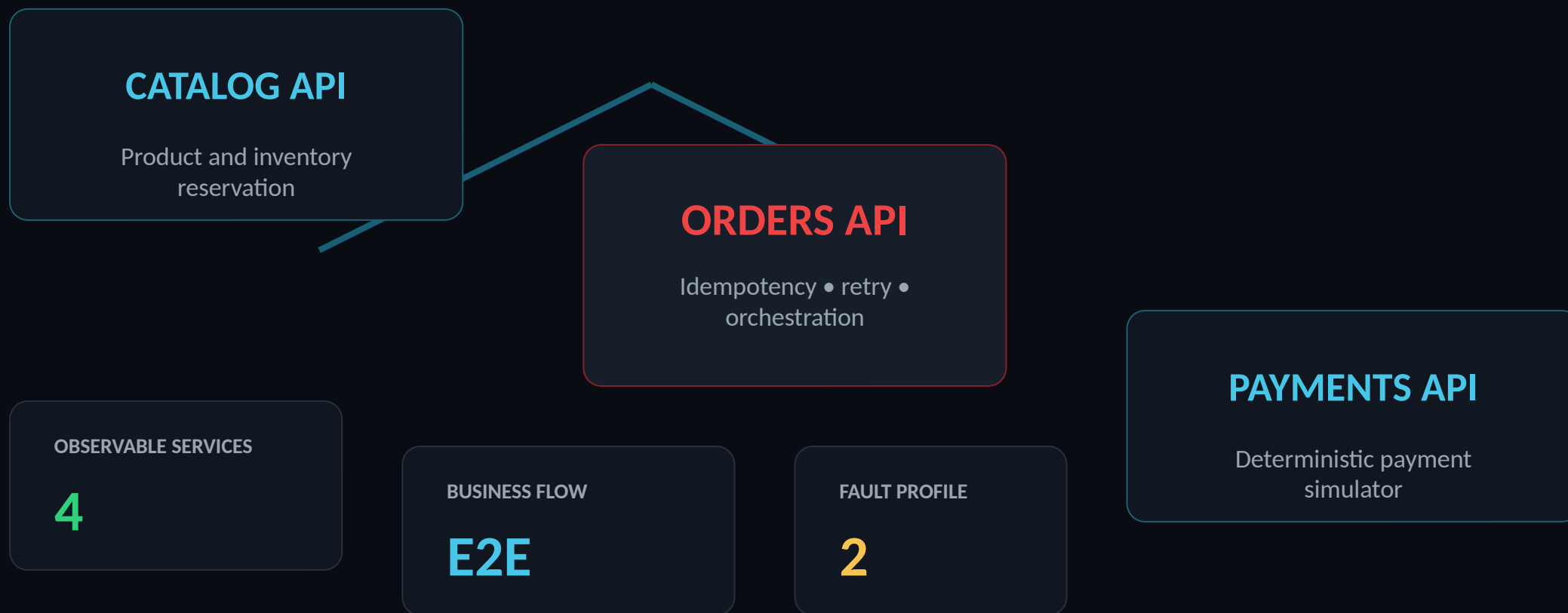


Core principle: an anomaly is not a cause; response begins only when evidence and guardrails agree.

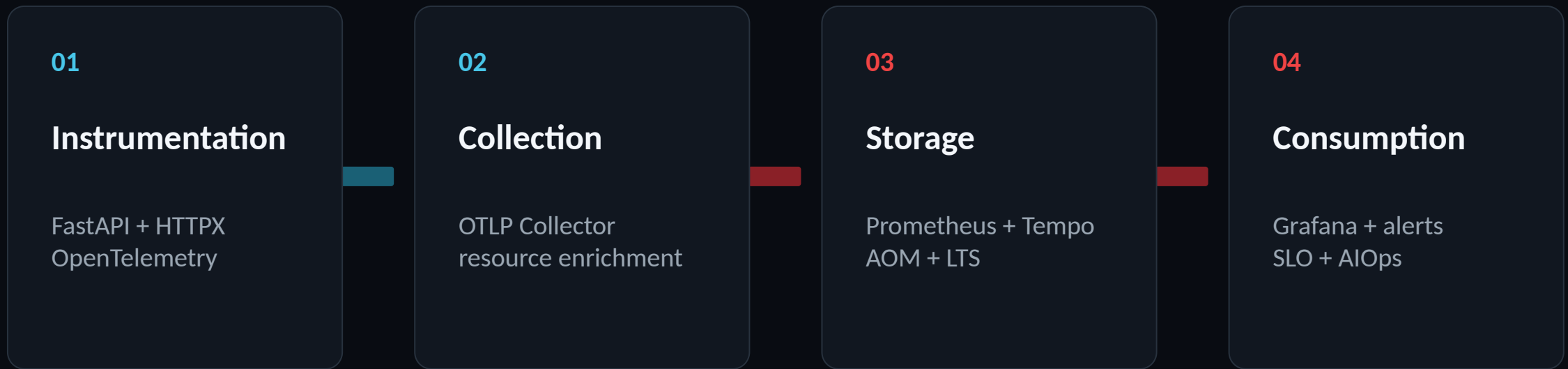
Four planes create a clear path from observation to safe response



The demo workload generates realistic operational behavior under control



Portable telemetry connects local tooling with Huawei Cloud operations



Application code stays backend-neutral; routing changes at the collector.

The hybrid model combines explainable statistics with pattern detection

65%

Robust deviation score

Median + MAD
Resistant to outliers
Human-readable evidence

**COMBINED
SCORE**

35%

Isolation Forest

Value + rate of change
Unsupervised pattern detection
Deterministic random state

Synthetic replay makes the critical departure from normal behavior visible



COMBINED SCORE

0.91

SEVERITY

CRITICAL

BASELINE MEDIAN

0.21

s

Root-cause ranking uses correlated evidence, not a single alert

1

CPU saturation and insufficient replicas

CPU + p95 latency



0.91

2

Database pool pressure

Pool + p95 latency

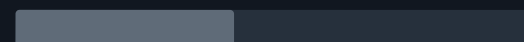


0.74

3

Unstable application release

Restarts + error rate



0.42

Incomplete evidence produces no hypothesis; low confidence blocks action.

SLOs and error budgets connect alerting to customer impact

99.9%

30-day availability objective

≈ 1,000 failed requests allowed per 1,000,000 requests

ERROR BUDGET



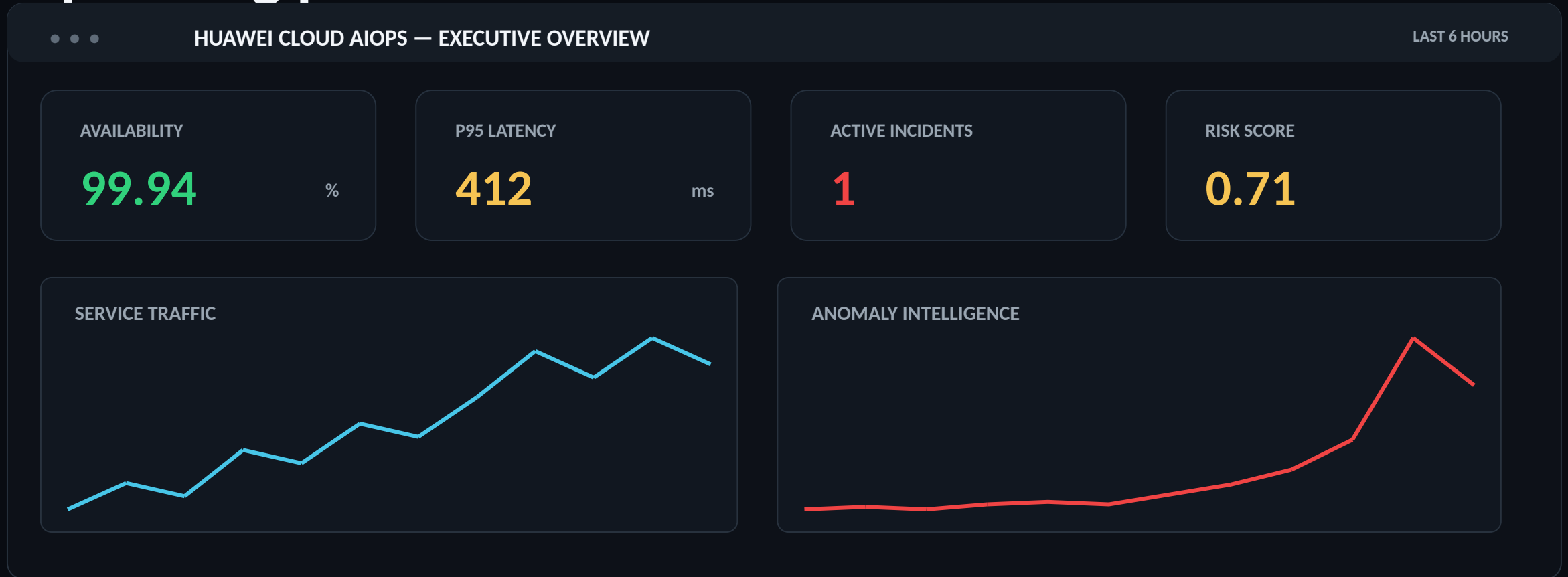
75% remaining

250 consumed

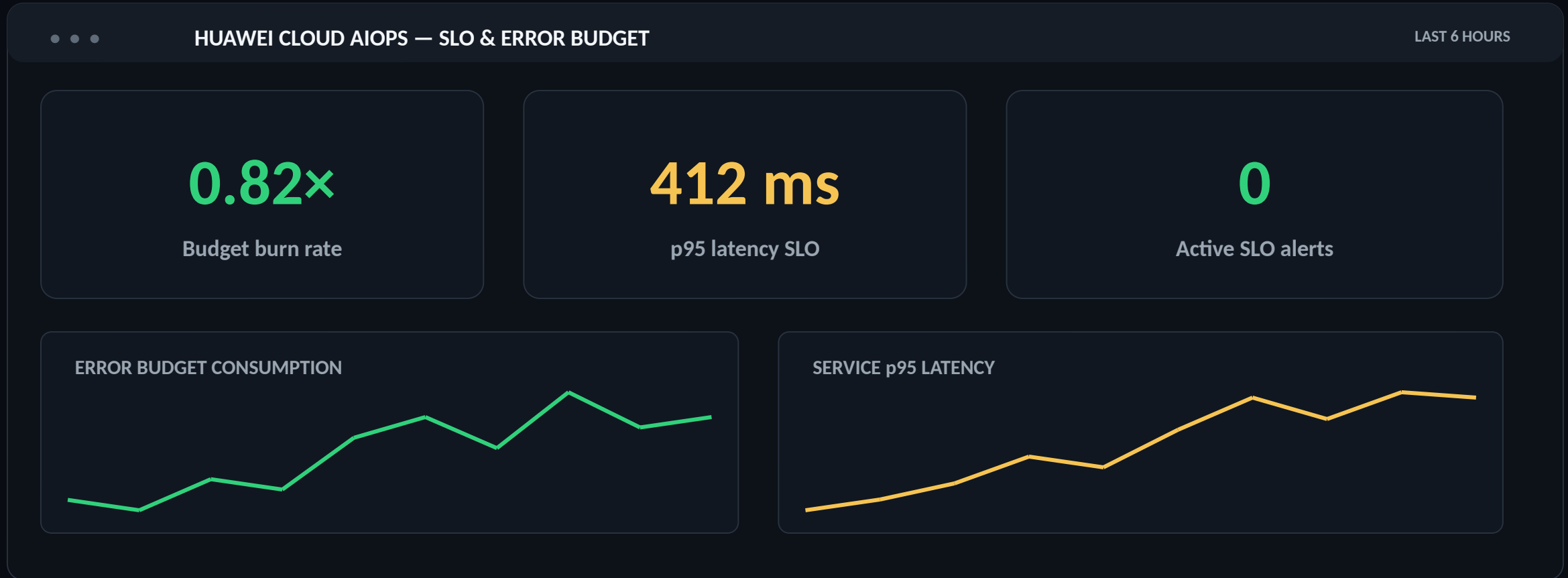
- 14.4× fast burn → critical
- 3× slow burn → warning

The alert asks whether reliability risk is urgent—not merely whether a metric moved.

The executive view compresses health and risk into one operating picture



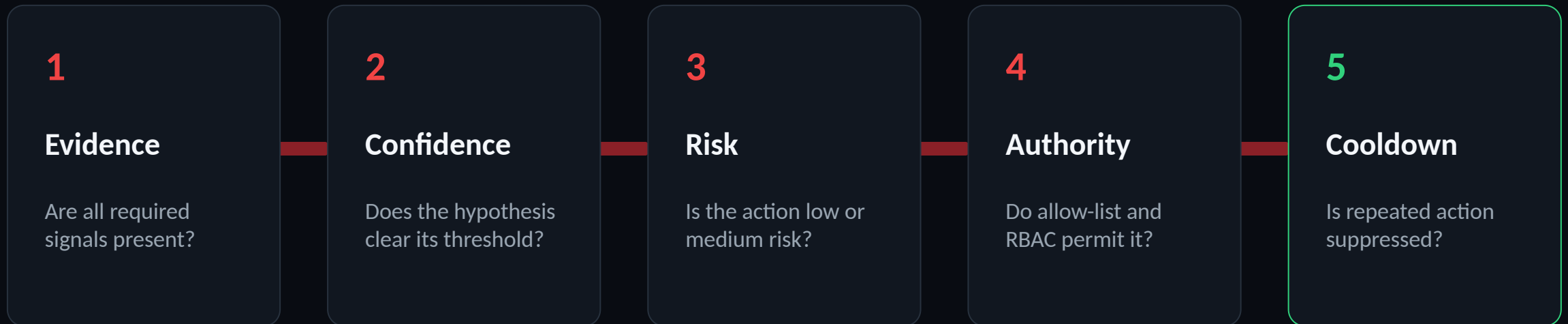
The SLO view tracks budget consumption across fast and slow burn windows



Incident intelligence links anomaly, action outcome, and business impact



Automation prioritizes reversibility and containment over speed



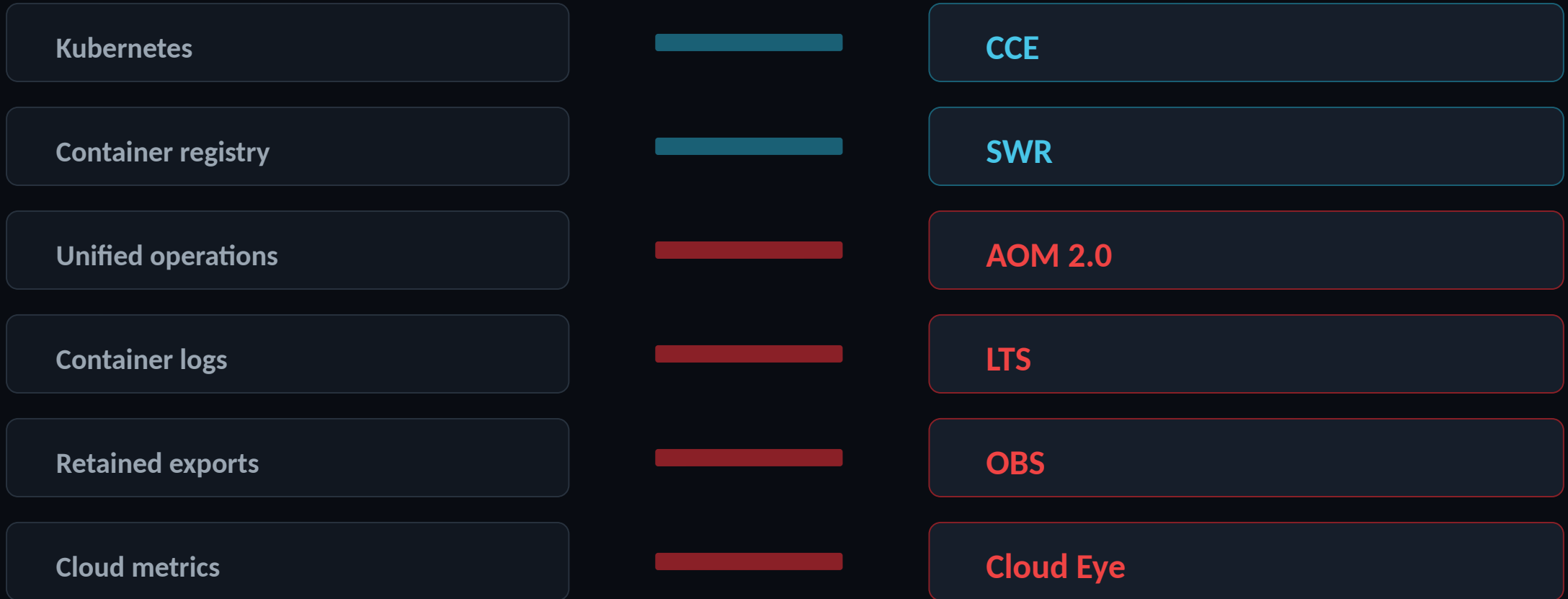
Rollback is high risk → it is recommended, never auto-executed.

One incident replay exercises the full chain from detection to verification



The goal is not a vague “self-healing” claim—it is measurable, auditable, safe automation.

Open components map cleanly to Huawei Cloud managed services



Status: cloud-deployment-ready • real apply requires an authorized account and approval

Defense layers deliberately minimize the automation blast radius

Supply chain

Trivy 0.35.0 • CodeQL v4 • pinned dependencies

Container

non-root • read-only FS • no privilege escalation

Kubernetes

default-deny NetworkPolicy • HPA • PDB • probes

Automation

namespace Role • allow-list • cooldown
• audit

Delivery

private SWR •
immutable
tag/digest • manual
cloud publish

The validation suite measures quality from code to executive delivery

PASSING TESTS

36

CODE COVERAGE

93.76

%

GRAFANA PANELS

22

RUNBOOK

6



Ruff lint + format



Docker Compose config



Terraform fmt + validate



Pytest + branch coverage



Helm lint + render



Trivy + CodeQL

The deliverable set supports both technical review and executive discussion

APPLICATION

4 FastAPI services + AIOps engine

OPERATIONS

3 Grafana dashboards + SLO + alerts

CLOUD

Terraform + Helm + CCE/SWR/OBS

SECURITY

RBAC + NetworkPolicy + Trivy + CodeQL

DOCUMENTATION

6 runbooks + 4 ADRs + threat model

EXECUTIVE

20-slide TR + 20-slide EN + executive PDF

The repository presents an operating model and decision story—not code alone.

Outcome: faster detection, clearer diagnosis, safer response

01

VISIBILITY

Metrics, logs, traces, and business signals in context

02

INTELLIGENCE

Explainable hybrid anomaly and root-cause ranking

03

CONTROL

Dry-run, RBAC, risk thresholds, cooldown, and audit

Next step: controlled cloud plan → dry-run evidence → approved activation